Introduction to Machine Learning Problems Unit 2: Simple Linear Regression

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1. (a) A possible target variable is the students' GPA, as it can be normalized on a scale between 0 and 100.

   (b) The target variable is continuous-valued.

   (c) The students' previous academic GPA could be a predictor for their current GPA.

   (d) A linear model would be reasonable; GPA would directly correlate with the success of a student. Outliers, such as vastly different majors or colleges, would exist, but would not have a severe impact on the linearity of the model.

2. (a) $\overline{x} = 2$ and $\overline{y} = 6$

   (b) $s_x^2 = 2.5$, $s_y^2 = 46.5$, and $s_{xy} = 10$

   (c) $\beta_0 = -2$, $\beta_1 = 4$, and $RSS(\beta_0, \beta_1) = 26$

   (d) $f(x = 2.5) = 8$

3. (a)
$$z(t) \approx z_0 e^{-\alpha t} \tag{1}$$
$$ln(z(t)) \approx ln(z_0 e^{-\alpha t}) = ln(z_0) + -\alpha t \tag{2}$$
   z_0 is a constant, and taking the log of the constant results in a constant: $ln(z_0)$ is our intercept and $-\alpha$ is our slope.

   (b) The LS formula is defined as $y = \beta_0 + \beta_1 x$; ergo, from **Eq.2**, $\beta_0 = ln(z_0)$ and $\beta_1 = -\alpha$

   (c)
```
import numpy as np

y = np.log(z)

beta_0, beta_1 = np.polyfit(t, y, 1)
alpha = -beta_0
z_0 = np.exp(beta_1)
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4. (a)

$$RSS(\beta) = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{3}$$

Let $\hat{y}_i = \beta x_i$, then substituting into **Eq.2** results in

$$RSS(\beta) = \sum_{i=1}^{n}(y_i - \beta x_i)^2 \tag{4}$$

(b)

$$\frac{dRSS(\beta)}{d\beta} = \frac{d}{d\beta}\sum_{i=1}^{n}(y_i - \beta x_i)^2 \tag{5}$$

$$= 2 * \sum_{i=1}^{n}(y_i - \beta x_i)(-x_i) \tag{6}$$

$$= -2 * \sum_{i=1}^{n}(x_i y_i - \beta x_i^2) \tag{7}$$

Setting the equation to 0 results in:

$$-2 * \sum_{i=1}^{n}(x_i y_i - \beta x_i^2) = 0 \tag{8}$$

$$\sum_{i=1}^{n}(x_i y_i - \beta x_i^2) = \tag{9}$$

$$\sum_{i=1}^{n} x_i y_i = \beta \sum_{i=1}^{n} x_i^2 \tag{10}$$

Solving for β results in

$$\beta = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}$$